App Development Project Report

App Dev Project Report

1. Student Details

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About Me: I am a student at IIT Madras BS Degree program with interest in application development. I like building applications for learning and those that support user experience and solve problems.

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2. Project Details

Project Title: Placement Portal Application

Problem Statement:

To build a Placement Portal Application web application that allows Admin (Institute), Company, and students to interact with the system based on their roles.

Approach:

The application was built using a role-based access control architecture. The backend logic is managed by Flask and session management. SQLite database is managed by SQLAlchemy. It provides an easy way to select, reject and shortlist students for companies. It also gives clear information to students about available placement drives, companies and status of their application. Admin can supervise the process and also select or reject companies and students when required. Implemented password hashing for security.

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3. AI/LLM Declaration

I used ChatGPT (GPT-5) and Gemini while making this project. The total usage is around 20%. I used it for database syntax, specific regular patterns for form validation, Chart.js integration, some Bootstrap 5 layouts, SQLAlchemy unique constraints and file validation for secure resume uploads.
The system architecture, database design, backend routing, and the integration of student, admin, and company were done manually.

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4. Technologies and Frameworks Used

Technology / Library	Purpose
Flask	Core backend web framework
SQLAlchemy	ORM for managing users, profile, and drives
Jinja2	Template for role based dashboards
Bootstrap 5	Responsive UI design and layout
Chart.js	To visualize placement statistics, student applications and application status
Werkzeug	Secure file handling for student resume uploads
SQLite	Relational database for storage

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5. Database Schema / ER Diagram

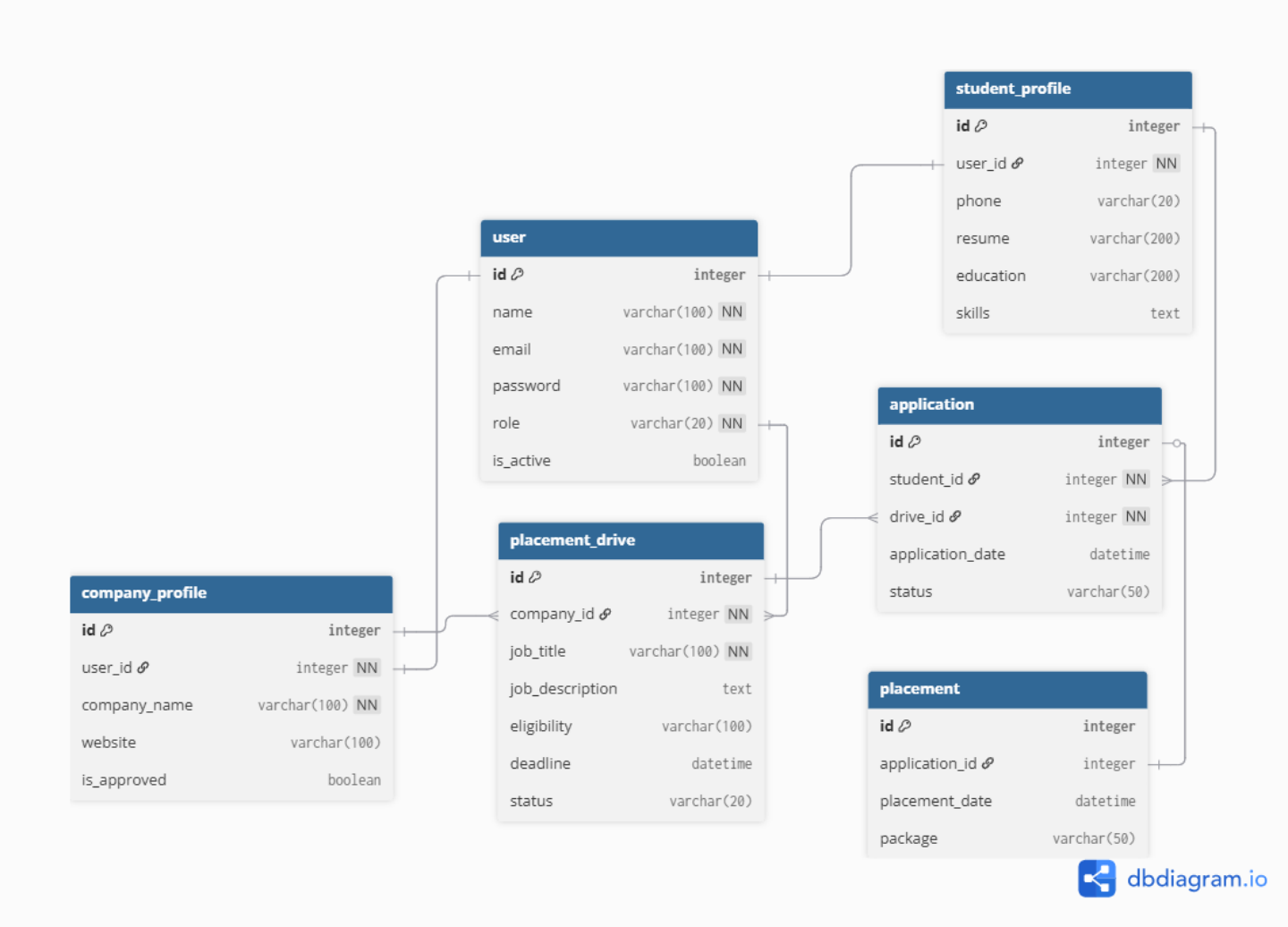
Tables:

1. **User:** Primary account table (id, name, email, password, role, is_active).
2. **StudentProfile:** Extended student data (id, user_id, phone, skills, resume).
3. **CompanyProfile:** Extended company data (id, user_id, company_name, is_approved).
4. **PlacementDrive:** Job posting details (id, company_id, job_title, eligibility, status).
5. **Application:** Linking students to drives (id, student_id, drive_id, status).

Relationships:

- **One-to-One:** User ↔ StudentProfile / CompanyProfile.
- **One-to-Many:** CompanyProfile → PlacementDrive.
- **One-to-Many:** PlacementDrive → Application.
- **One-to-Many:** StudentProfile → Application.

ER Diagram:



(taken from dbdiagram.io)

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6. API Resource Endpoints

Endpoint	Method	Description
/api/jobs	GET	Retrieves all currently active placement drives and eligibility criteria.
/api/student/<id>/applications	GET	Fetch application history and statuses for specific student ID.
/api/apply	POST	Submits JSON payload to create a new application record.

/api/application/<id>	PUT	Updates the status of an existing application.

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7. Architecture and Features (optional)

Architecture Overview:

- **app.py:** Main entry point containing all route controllers.
- **model.py:** SQLAlchemy class definitions and database relationships.
- **/static/resumes:** Secure storage for uploaded PDF/Docx resumes.
- **/static/css/style.css:** Custom CSS overrides for Bootstrap.

Implemented Features:

- **Role based authentication:** Separate login for Admin, Company, and Student.
- **Approval of admin:** Admin must approve companies before they can post jobs.
- **Resume Management:** Students can upload, view, and delete their resumes.
- **Search:** Admin can search students and companies by name or ID.
- **Responsive UI:** Fully functional on mobile and desktop via Bootstrap 5.
- **Integrity :** Database constraints to prevent duplicate applications.

Additional Features:

- Added Chart.js to provide graphical insights.

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8. Video Presentation

Drive Link:

https://drive.google.com/file/d/1o4TZQ9fhNIF2YIWttrVUIzYcvcSzuQRA/view?usp=sharing
(Accessible to all with “View” permission.)
